

ENSF 380

TOPIC 05: TESTING (PART I)

Why Test Your Code?

Write better code

Better design

Identify and think about ambiguities

How to Test Your Code

What can go wrong?

```
// Take two integers and return the difference between them  
public static int differenceNumbers(int first, int second) {  
}
```

What if 'second' is bigger than 'first'?

What if both numbers are the same?

What if one number is negative?

What if both numbers are negative?

Writing a `main()` to Test Your Code

01_Main

```
public static void main(String[] args) {  
    // what if 'second' is bigger than 'first'?  
    System.out.print("2, 3 should give 1. Gave: ");  
    System.out.println(differenceNumbers(2,3));  
  
    // what if both numbers are the same?  
    System.out.print("34, 34 should give 0. Gave: ");  
    System.out.println(differenceNumbers(34,34));  
  
    // what if one number is negative? Is it seen as smaller?  
    System.out.print("-1, 2 should give 3. Gave: ");  
    System.out.println(differenceNumbers(-1,2));  
  
    // what if both numbers are negative? Do we get the right answer?  
    System.out.print("-1, -2 should give 1. Gave: ");  
    System.out.println(differenceNumbers(-1,-2));  
}
```

Exercise 5.1

TEST YOUR UNDERSTANDING BY COMPLETING THE EXERCISE

Precise Requirements? Sometimes.

What does this mean?

```
// Subtract the product from the sum unless it isn't positive,  
// in which case subtract the sum from the product but if it is -8 just  
// return zero
```

- **How many numbers are given as input?**
- **What type of variables does the method accept and return (int, double, float...)?**
- **If the result is neither negative nor positive (zero) what should it do?**
- **Does it return 0 if the sum minus the product is below -8, or only if the product minus the sum is?**
- **Is the return 0 condition triggered at -8 exactly, or -8 or lower (or higher)?**

```
// Take two integers. Subtract the product from the sum. Return  
// this number if it is positive. If it is below -8, return 0.  
// Otherwise, subtract the sum from the product and return that value.  
public static int subtractProductSum(int first, int second) { }
```

Edge Cases and Boundary Conditions

Edge case: a situation that occurs at extreme or unusual operating parameters

- Test for edge cases by looking for unusual or extreme situations ("what if?")

Boundary condition: the point at which different behavior is expected depending upon input

- Test a boundary condition by examining both sides of the boundary, as close to the boundary as possible

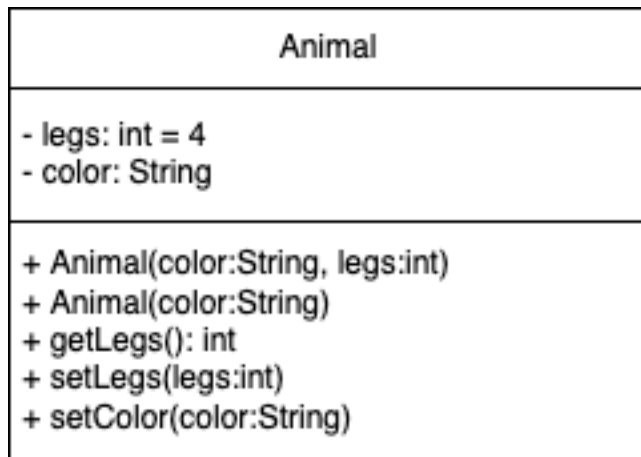
Exercise 5.2

TEST YOUR UNDERSTANDING BY COMPLETING THE EXERCISE

How We Grade

Example requirements

Create a file which implements the class shown in the UML class diagram.



Example questions

Is there a constructor that takes just a colour and uses the default (4) for the number of legs?

- Can we change the number of legs?

Is there a constructor that takes the colour and number of legs?

- Is an Animal object created?
- Was the number of legs set correctly?
- Can we change the number of legs?
- Is color a private variable?

...

How We Grade

Example questions

Is there a constructor that takes just a colour and uses the default (4) for the number of legs?

- Can we change the number of legs?

Is there a constructor that takes the colour and number of legs?

- Is an `Animal` object created?
- Was the number of legs set correctly?
- Can we change the number of legs?
- Is `color` a private variable?

...

Example testable questions

Create an `Animal` using the `Animal("green")`

1. If we use `getLegs()`, does it have the default number of legs (4)?
2. If we use `setLegs(10)`, does a subsequent call to `getLegs()` return 10?

Create an `Animal` using
`myAnimal = Animal("blue", 1)`

3. Is `myAnimal` an object belonging to the class `Animal`?
4. Does `getLegs()` return 1?
5. If we use `setLegs(3)`, does a subsequent call to `getLegs()` return 3?
6. Does `myAnimal.color` give an access error?

How We Grade

Important: All code submitted in this class is expected to compile. Code which doesn't compile will score zero points.

Example testable questions

Create an Animal using the `Animal("green")`

1. If we use `getLegs()`, does it have the default number of legs (4)?
2. If we use `setLegs(10)`, does a subsequent call to `getLegs()` return 10?

Create an Animal using
`myAnimal = Animal("blue", 1)`

3. Is `myAnimal` an object belonging to the class `Animal`?
4. Does `getLegs()` return 1?
5. If we use `setLegs(3)`, does a subsequent call to `getLegs()` return 3?
6. Does `myAnimal.color` give an access error?

Implications

Errors can have a cascading impact

If `setLegs()` doesn't work, two tests will fail (2, 5); reason is irrelevant

- Incorrect implementation
- Method not implemented (`setLeg()` means `setLegs()` isn't implemented; `setlegs()` means `setLegs()` isn't implemented)

If `getLegs()` doesn't work, four tests will fail (1, 2, 4, 5)

If there is no constructor that takes two arguments, four tests will fail (3-6)

Correctness is evaluated by the compiler & tests

How to Interpret Test Feedback

Feedback	Meaning	
testStringOnlyConstructor: Default legs number should be 4 - expected:<4> but was:<0>	Test name (condition tested)	testStringOnlyConstructor
	Test description (expected outcome)	Default legs number should be 4
	Test expected the value	4
	Test received the value	0

Interpretation: When constructing the object with a single argument, the number of legs should be set to the default, which is 4. However, the number of legs was set to 0.

Exercise 5.3

TEST YOUR UNDERSTANDING BY COMPLETING THE EXERCISE

Why We Use Testing to Grade

Industry practice involves using tests to gauge correctness of code

TA time can be spent on interactive help

Consistency

Demo

HOW WE GRADE

Tips for Getting Good Marks on Assessments

Ensure your code compiles

Think about edge cases and test for them

Ask questions (discussion board, labs) if requirements are unclear

Pay particular attention to getters, setters, and constructors

Ensure there are no errors in the submission format

Use the grace period as intended; no late submissions

Why Debugging

Be able to fix and complete your own code

Practice code maintenance

Debugging with RIMGEN

Replicate

Isolate

Maximize

Generalize

Externalize

Neutral tone

Generalizing

Example: a method takes a question as input and returns an answer.

Input	Output
How are you?	I'm good.
Wht are you?	I'm a miserable piece of code.
But you just siad you were good?	I'm bad.
How can that be?	It's part of the beautiful mysteries of the universe.
What is goinon?	In an indifferent universe, anything can happen.
What is going on?	It's all good.

Rule: If there is a spelling error in the sentence, the response will be negative. Otherwise it will be positive.

Exercise 5.4

TEST YOUR UNDERSTANDING BY COMPLETING THE EXERCISE

Summary: Knowledge

Why you should test your code

How to move from requirements to questions to testable questions

Edge cases and boundary conditions

How and why your assessments are graded with tests in this class; what testing means for grading

How to interpret test feedback

The RIMGGEN debugging framework

Summary: Skills

Writing a `main()` to test your code

Debugging using "Replicate, Isolate, Maximize, Generalize"